alzheimer mri stage classification

using cnn and googlenet-like cnn

deep learning project report

dataset	Falah/Alzheimer_MRI
task	medical image classification
models	simple cnn, googlenet-like cnn
best validation accuracy	91.60%

note: this system provides computer-aided preliminary classification only and is not a final medical diagnosis.

1. project idea

This project classifies brain MRI images into Alzheimer-related stages using convolutional neural networks. The model receives one MRI image and predicts one of four classes: mild_demented, moderate_demented, non_demented, and very_mild_demented.

The project is designed as a medical image classification system. The prediction is a preliminary computer-aided classification, not a final medical diagnosis.

2. dataset

The dataset used is Falah/Alzheimer_MRI from Hugging Face. It contains brain MRI images divided into four classes.

class index	class name
0	mild_demented
1	moderate_demented
2	non_demented
3	very_mild_demented

3. preprocessing

The preprocessing pipeline follows the image classification layout used in the deep learning sheets: transform, dataset, dataloader, model, loss, optimizer, training, validation, and testing.

step	purpose
resize to 64 x 64	unifies image size before entering the cnn
grayscale to 3 channels	keeps compatibility with conv2d input channels
totensor	converts image to pytorch tensor
normalization	stabilizes pixel value range for training

4. model architectures

4.1 simple cnn

The simple cnn was used as the baseline model. It consists of two convolution blocks followed by fully connected layers.

Architecture: conv2d -> relu -> maxpool -> conv2d -> relu -> maxpool -> flatten -> linear -> relu -> output layer.

4.2 googlenet-like cnn

The googlenet-like cnn uses inception-style blocks. Each block applies multiple convolution branches and concatenates their outputs. This helps the model learn different levels of image features from MRI images.

Architecture: conv2d -> relu -> maxpool -> inception block -> maxpool -> inception block -> maxpool -> flatten -> linear -> relu -> output layer.

5. training setup

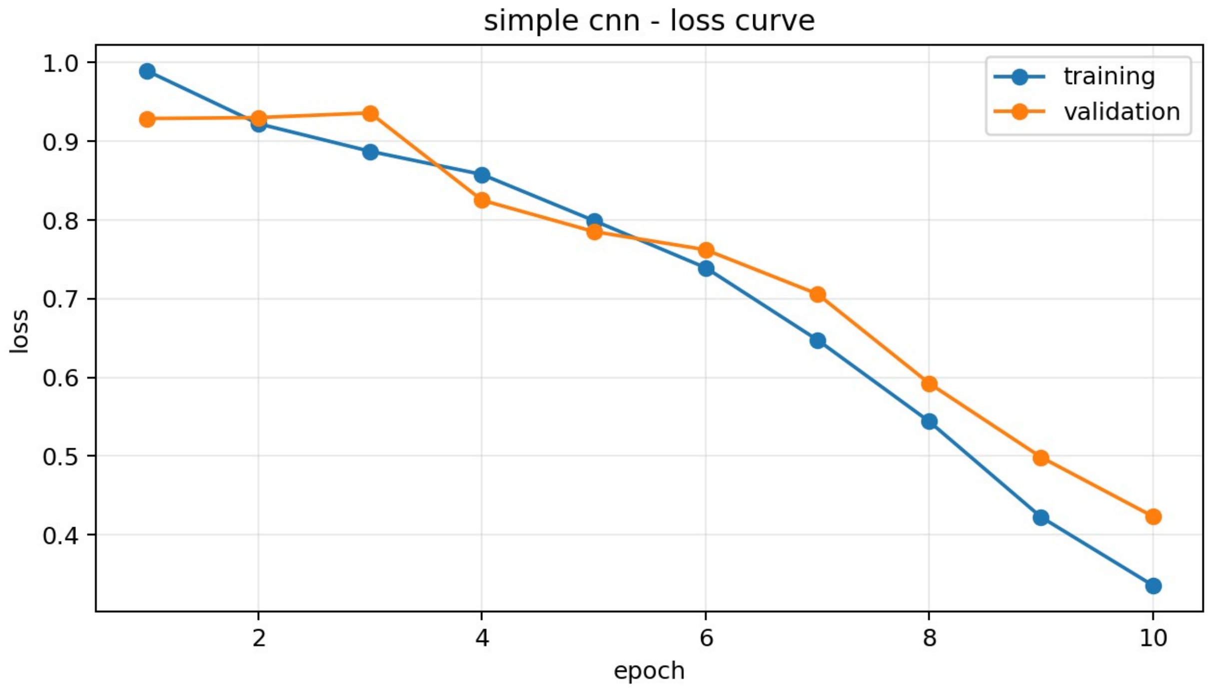
item	value
loss function	CrossEntropyLoss
optimizer	Adam
number of epochs	10
number of classes	4
evaluation during training	training loss, training accuracy, validation loss, validation accuracy

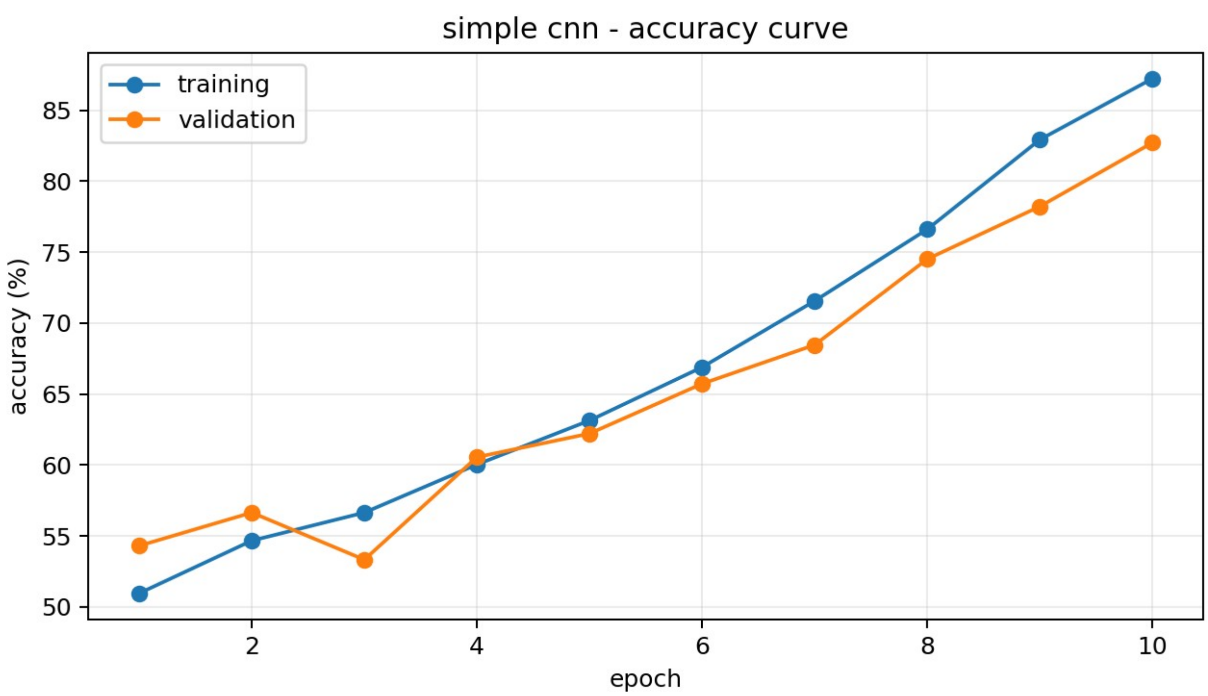
6. training results

6.1 simple cnn results

epoch	train loss	train acc	val loss	val acc
1	0.9899	50.95%	0.9290	54.30%
2	0.9224	54.66%	0.9302	56.64%
3	0.8873	56.64%	0.9361	53.32%
4	0.8579	60.03%	0.8252	60.55%
5	0.7993	63.13%	0.7853	62.21%
6	0.7394	66.89%	0.7622	65.72%
7	0.6476	71.56%	0.7060	68.46%
8	0.5439	76.61%	0.5924	74.51%
9	0.4224	82.93%	0.4985	78.22%
10	0.3356	87.23%	0.4233	82.71%

6.1 simple cnn curves

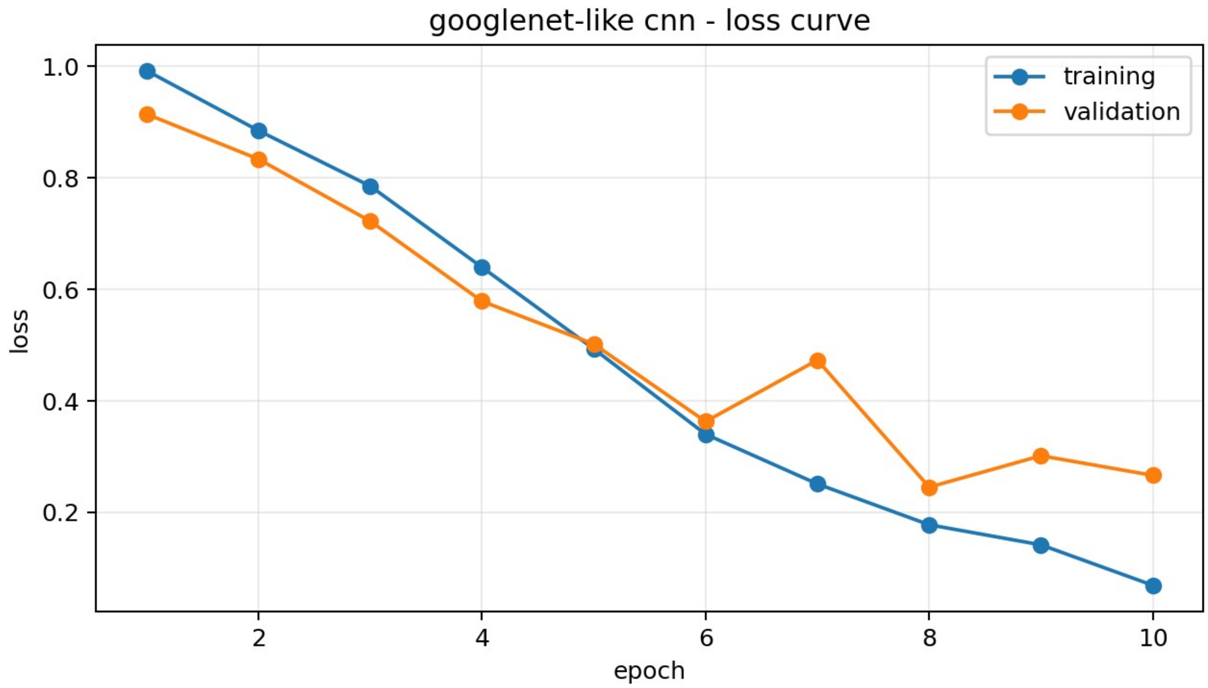


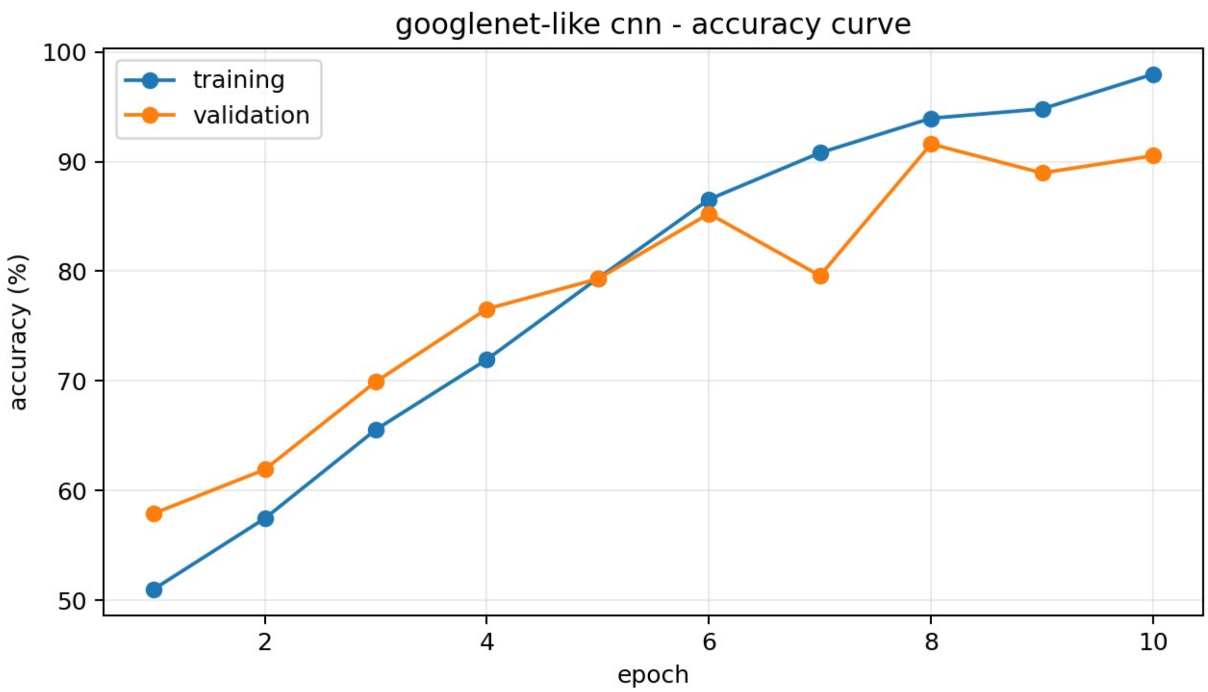


6.2 googlenet-like cnn results

epoch	train loss	train acc	val loss	val acc
1	0.9926	51.00%	0.9147	57.91%
2	0.8851	57.45%	0.8339	61.91%
3	0.7855	65.55%	0.7226	69.92%
4	0.6398	71.95%	0.5788	76.56%
5	0.4933	79.35%	0.5020	79.30%
6	0.3402	86.55%	0.3630	85.25%
7	0.2510	90.80%	0.4732	79.59%
8	0.1777	93.95%	0.2452	91.60%
9	0.1417	94.80%	0.3016	88.96%
10	0.0691	97.97%	0.2664	90.53%

6.2 googlenet-like cnn curves





7. comparison table

model	final train acc	final val acc	final train loss	final val loss
simple cnn	87.23%	82.71%	0.3356	0.4233
googlenet-like cnn	97.97%	90.53%	0.0691	0.2664

model	best val acc	epoch
simple cnn	82.71%	10
googlenet-like cnn	91.60%	8

8. result analysis

The simple cnn improved steadily. Its training accuracy increased from 50.95% to 87.23%, and its validation accuracy increased from 54.30% to 82.71%.

The googlenet-like cnn achieved stronger validation performance. Its final validation accuracy was 90.53%, and its best validation accuracy was 91.60% at epoch 8.

The googlenet-like cnn showed a small sign of overfitting after epoch 8, because training accuracy continued to increase while validation accuracy decreased slightly. However, it still achieved the best overall validation performance.

9. conclusion

The project successfully implemented Alzheimer MRI image classification using two cnn-based models: simple cnn and googlenet-like cnn.

The googlenet-like cnn performed better than the simple cnn. The best validation accuracy achieved was 91.60%.

The final saved models can be used in the streamlit deployment for single image prediction and batch prediction with dataset distribution summary.